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MAHKT: Knowledge tracing with multi-association heterogeneous graph embedding based on knowledge transfer

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ABSTRACT

Knowledge Tracing (KT) is crucial in online education for tracing learners knowledge states and improving learning outcomes. However, existing graph-based KT models often focus solely on prerequisite associations among concepts or correlations between concepts and questions, neglecting the complex multi-association relationships among entities. This oversight causes biased question representations, which in turn affects the knowledge transfer propagation process, ultimately leading to inconsistencies between the represented and actual knowledge states. To address this, we propose multi-association heterogeneous knowledge transfer (MAHKT), a novel KT model that leverages multi-association heterogeneous graph embeddings based on knowledge-transfer principles, to better capture the evolution of knowledge states. Five types of associations were defined between questions and concepts to construct a multi-association heterogeneous graph, thereby proposing a knowledge triplet embedding method to extract the potential value of these associations. Grounded in near- and far-transfer theories, MAHKT employs a perception-attention mechanism to calculate similarity weights, propagates embeddings from first-order neighbors to capture near-transfer effects, and stacks propagation layers to aggregate far-transfer information, thereby comprehensively integrating richer knowledge from similar knowledge transfers across various association contexts. Finally, two LSTM modules were utilized to track the evolution of learner's knowledge states across questions and concepts, to predict their future performance. The experimental results show that MAHKT significantly outperformed existing KT methods, enhancing prediction accuracy and improving model interpretability in the knowledge transfer process.

1. Introduction

In recent years, the rapid growth and widespread adoption of online education platforms have transformed learning, offering significant convenience to teachers and learners [1–3]. With the rise of artificial intelligence, these platforms now enable intelligent resource recommendations [4] and personalized teaching interventions [5], such as intelligent tutoring systems and massive open online courses, making them a crucial part of modern education [6].

Knowledge tracing (KT) is a crucial technology that has been widely integrated into online education platforms in data-driven intelligent education. The primary aim of KT is to effectively monitor learners' knowledge states by analyzing the temporal response data from their learning processes, enabling the prediction of their future performance, as illustrated in Fig. 1(a). Researchers have proposed various KT approaches, among which Bayesian knowledge tracing (BKT) [7] employs a hidden Markov model to trace the evolving knowledge state of each concept. Factor analysis [8,9] predefines a delicate model framework and assigns the considered factors as model parameters, which are learned from the data to generalize the observations. Deep knowledge tracing (DKT) [10] uses recurrent neural networks to jointly model the states of all concepts. However, these classic KT models overlook the knowledge graph structure, leading to loss of entity information.

Recently, many researchers have recognized the importance of associations between questions and concepts [11]. Graph-based knowledge tracing (GKT) utilizes prerequisite associations among concepts to construct a concept graph. Structure-based knowledge tracing (SKT) [12]

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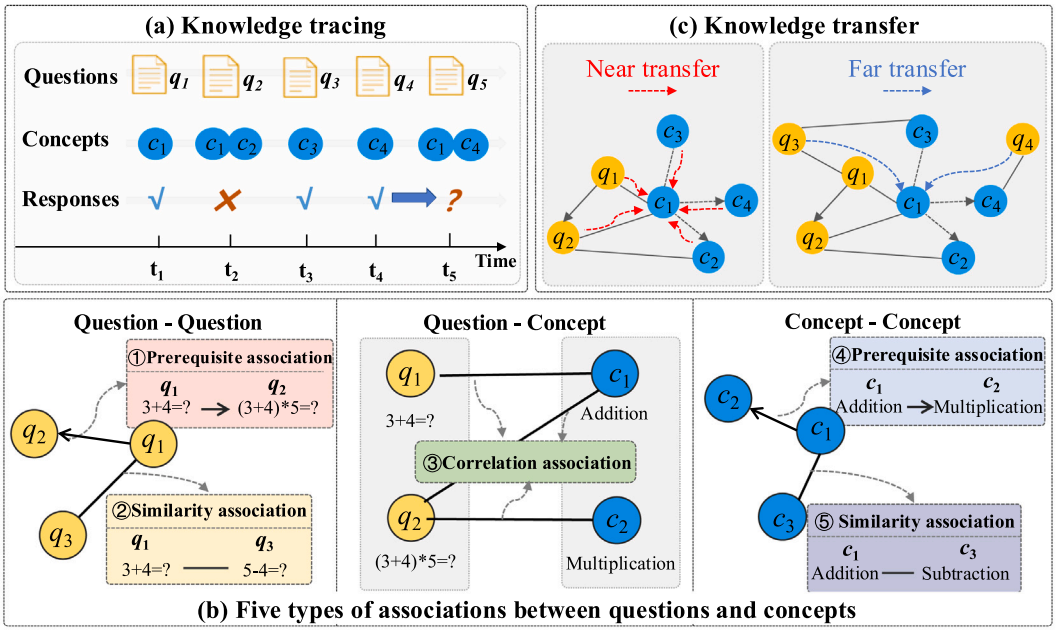


Fig. 1. A toy example. (a) Illustrates the knowledge tracing process; (b) Shows the five types of associations between questions and concepts: the left diagram depicts associations among questions, including prerequisites and similarities; the middle diagram shows correlations between questions and concepts; and the right diagram depicts associations among concepts, including prerequisites and similarities. (c) Illustrates the near and far transfer processes in knowledge propagation.

utilizes similarity and prerequisite associations within concepts to enrich representation, and multi-relation enhanced graph-based knowledge tracing (MGKT) [13] explores similarity associations among questions through constructing graph structure with hypergraph reconstruction. These models are rooted in graph neural networks and have undergone substantial improvements and refinements. However, they focus solely on prerequisite associations among concepts or correlations between concepts and questions, neglecting the complex multi-association relationships among entities. Such neglect would negatively affect the model's ability to effectively capture necessary relational information for accurately representing questions, leading to incomplete semantic features. Consequently, knowledge state representations derived from these question embeddings may become biased. Moreover, this oversight complicates the model's ability to update and aggregate knowledge information from related nodes during knowledge transfer. As a result, the inconsistency in knowledge state representations during transfer further negatively impacts the model's predictive performance.

While addressing the inconsistency in knowledge state representations is crucial for the KT task, two major challenges must be overcome to achieve this goal. On the one hand, the representation of questions depends on complex associations with concepts. In fact, as depicted in Fig. 1(b), the knowledge structure can be modeled as a heterogeneous graph with five types of associations: prerequisite and similarity associations among questions and among concepts, as well as correlation associations between questions and concepts. As shown in Fig. 1(b), the question $q_1(3+4=?)$ serves as a prerequisite for question $q_2((3+4)*5=?)$; q_1 also exhibits similarity to the question $q_3(5-4=?)$. Similarly, concept $c_1(Addition)$ acts as a prerequisite for concept $c_2(Multiplication)$; c_1 shows case similarity to concept $c_3(Subtraction)$. Moreover, a correlation exists between concept c_1 and question q_1. Mining these interrelated data can assist with the comprehension of association between questions and concepts, improving the performance of the KT model. However, inadequate association capture or integration can lead to deviations in the overall semantic features of the question representation. For instance, there is the similarity association between questions q_1 and q_3. When this similarity association is ignored, questions q_1 and q_3 may be represented as independent representation vectors, failing to reflect their inherent semantic connection. Such biased question representations hinder the model's ability to accurately trace knowledge states, particularly when representing similar questions, thereby reducing predictive accuracy. Therefore, mining and integrating these complex associations is crucial to improving the performance of KT methods.

On the other hand, knowledge propagation follows near- and far-transfer theories [14,15], which depends on the similarity and strength of associations between entities. According to the common elements theory [16], knowledge transfer is more likely when two entities share common features. The more the similarities, the higher is the likelihood of near transfer; when similarities are fewer, far transfer occurs [17]. Near transfer involves sharing information across similar domains through direct associations, such as correlation, similarity, or prerequisites. As shown in Fig. 1(c) (left part), concept c_1 is directly linked to concepts (c_2, c_3, c_4) and questions (q_1, q_2), necessitating the aggregation of these features. Near transfer is generally more common [18] than far transfer although the latter has been validated by many scholars [15,19,20]. Far transfer occurs across less related domains, where indirect associations exert a weaker influence. In Fig. 1(c) (right part), concept c_1 is indirectly linked to questions q_3 and q_4, indirectly influencing the current concept, which is the example of the far transfer. A comprehensive question representation must integrate these varied associations; however, developing effective methods for this purpose remains a challenge. When complex associations, such as prerequisite and similarity associations, are overlooked, the knowledge information of related nodes cannot be appropriately updated and aggregated during knowledge propagation. For instance, if the prerequisite association between concepts c_1 and c_4 is ignored, not only the direct adjacent prerequisite features cannot be aggregated effectively, but also the indirect association between c_1 and question q_4 cannot be captured. This will lead to inappropriate updating and aggregation of c_1 during both the near and far transfer processes, resulting in incomplete question representation. As a result, this exacerbates the inconsistencies in knowledge state updates and further degrades model performance.

Our research addresses these gaps by capturing all associations in a heterogeneous question graph through near and far transfers, overcoming the limitations of current models. We categorized these associations into five types: prerequisites and similarities among questions, prerequisites and similarities among concepts, and correlations between questions and concepts. Current methods often consider these associations

only partially, leading to inconsistencies in knowledge state representations during the transfer process. To address this issue, we introduce a novel approach called multi-association heterogeneous graph embedding based on knowledge transfer (MAHKT). The five associations were fully integrated into a multi-association heterogeneous graph, transforming the graph into knowledge triplets (entity, association, entity) and applying an embedding learning method to convert heterogeneous entities into dense vectors. MAHKT uses a perception-attention mechanism to calculate similarity weights, propagate embeddings from first-order neighbors for near transfer, and stack propagation layers for far-transfer aggregation. To predict future performance, MAHKT adopts a dual-LSTM approach to track the evolution of learner's knowledge states along both question and concept dimensions. The contributions of this study can be summarized as follows:

- The proposed MAHKT is a model that effectively captures implicit knowledge associations between questions and concepts, by constructing a multi-association heterogeneous graph that accounts for all five types of associations.
- To improve embeddings, a knowledge propagation method that captures near transfer is proposed through attention-based aggregation of neighboring nodes, further incorporating far transfer into stacked layers. Additionally, dual LSTM modules are used to trace learner's knowledge states for better performance predictions.
- Extensive experiments were conducted on real-world datasets to validate the effectiveness of MAHKT in terms of accuracy and interpretability.

The rest of the paper is organized as follows: Section 2 investigates and analyzes related works of our research. Section 3 presents a formal problem definition. Section 4 provides a detailed description of MAHKT. Section 5 presents and analyzes the extensive experiments implemented of MAHKT. Section 6 summarizes the major work of this paper.

2. Related work

2.1. Knowledge tracing

Currently, various deep learning methods have been applied to KT [21,22], with mainstream approaches falling into four categories. First, sequential neural network-based KT models such as learning process-consistent knowledge tracing (LPKT) [23] and hierarchical memory network (HMN) [24] use recurrent neural networks (RNNs) to model answering sequences of learners, with the hidden vectors representing learners' knowledge states. Second, key–value memory-network-based KT models, such as sequential key–value memory network (SKVMN) [25] and cognitive features dynamic knowledge diagnosis (CF-DKD) [26], extend DKT by capturing knowledge states at each moment to improve question representation. Third, attention-based KT models such as adversarial training-based knowledge tracing (ATKT) [27] and fine-grained knowledge tracing (FKT) [28] use masked attention to simulate the evolution of learners' knowledge states, thereby attracting attention for their excellent performance. In addition, to better describe the associations between questions and concepts, many researchers have proposed using graph neural networks, as in models such as deep graph memory networks (DGMN) [29] and heterogeneous graph-based temporospatial knowledge tracing (TSKT) [30], which offer better association interpretability and excellent performance.

2.2. Graph representation learning

Graph representation learning methods such as graph neural networks (GNNs) have proven successful in dealing with graph-structured data across a variety of applications. GNNs excel in extracting and encoding high-dimensional non-Euclidean information from graphs, unlike standard machine learning methods. GNNs produce node representations that capture both local graph neighborhood structure and placement within the overall graph [31] by propagating and aggregating information from surrounding nodes based on graph edges. These graph-based representations can be used directly for various downstream tasks [32]. Deep learning models using GNNs solve the sparsity issue associated with one-hot representations in the context of KT tasks. A commonly used GNN model is the graph convolutional network (GCN), which has been employed to represent learners' answers to questions in recent studies [33,34]. Heterogeneous graphs [35], as the graph with multiple types of vertices and edges, can effectively extract complex associations and rich semantic information from various data mining tasks. And many heterogeneous graph neural networks (HGNN) have been proposed and used in different research fields, such as the social recommendation [36], finance fraud detection [37], and information cascade prediction [38]. Furthermore, triplet embedding has emerged as a powerful technique in the graph-representation learning paradigm, particularly for heterogeneous graph representations [39]. This approach is highly suitable for KT tasks because the multi-association graph consisting of heterogeneous questions is a typical example of a heterogeneous graph. In this study, knowledge triplet embedding is leveraged to derive dense embedding vectors from graphs. These embedding vectors are then fed into knowledge propagation to enhance the embedding vectors.

2.3. Graph-based knowledge tracing

In various graph-based KT models, associations between questions and concepts have been considered [40]. For example, the pre-training embedding bipartite graph (PEBG) [41] establishes a bipartite graph of questions and concepts based on the Q matrix. By incorporating constraints such as question–concept correlation, concept similarity, and question similarity, pre-training is performed on the question and concept embedding maps. The graph-based interaction model for knowledge tracing (GIKT) [34] utilizes GCNs to integrate question–concept correlations through embedding propagation. SKT [12] utilizes multiple concept associations to influence propagation and handle similarities through simultaneous propagation, whereas prerequisites are managed through partial propagation. GKT [33] represents concept associations as graphs and redefines the KT task as a time-series node-level classification problem using GNNs. Graph-based attentive structure knowledge tracing (GASKT) [42] divides questions and concepts into two types of nodes and uses relational graph convolutional networks (R-GCNs) to integrate question–concept correlations through embedding propagation. Knowledge structure-enhanced graph-based attention knowledge tracing (KSGKT) [43] dynamically traces learners' knowledge proficiency by incorporating correlations and prerequisite associations between questions and concepts into the KT model. Joint graph convolutional knowledge tracing (JKT) [44] divides multidimensional associations between questions and concepts into isomorphic subgraphs for GCN training, capturing high-level semantic information and enhancing model interpretability. There have also been graph-based KT methods that attempt to capture the latent associations between questions and concepts in other ways, such as enriching node types through heterogeneous graphs or adopting multi-view enhancement methods. The heterogeneous graph-based programming knowledge tracing (HGPKT) [45] proposes a novel heterogeneous graph representation method for the KT task, which captures associations between knowledge concepts, programming questions, and corresponding codes using graph convolution and attention mechanisms for more accurate knowledge state tracing. The bi-graph co-contrastive learning KT model (BiCo) [46] proposes a novel question embedding pre-training method, which enhances the association extraction and embedding

Table 1
Notations and explanations.

Notations	Description	Notations	Descriptions
q	Question	e_r	Embedding of association
c	Concept	$\tilde{e}$	Enhanced embedding of the question
r	Association	v	Embedding of the fusion knowledge
n	Number of concepts	a	Learner's true response
m	Number of questions	x	Learner's interaction embedding
k	Number of associations	h_t^c	Concept-level knowledge state at step t
seq	Learner response sequence length	h_t^q	Question-level knowledge state at step t
e_h, e_t	Embedding of question or concept	$\hat{a}$	Predicted of learner response

representation of questions through multi-view semantic representations and structure recovery. The dual graph ensemble learning-based knowledge tracing (DGEKT) [47] captures heterogeneous exercise–concept associations and interaction transitions through hypergraph and directed graph modeling, and combines the dual graph models using knowledge distillation to enhance predictive performance. These methods indicate the significant impact of adequately capturing the complex associations between questions and concepts.

3. Problem formulation

Suppose that each question is associated with one or more concepts. Learners acquire these concepts by responding to questions, generating interactive record tuples, denoted as $x_t = (q_t, a_t)$. Each tuple includes the question $q_t \in \{q_1, q_2, \ldots, q_m\}$ and answer correctness $a_t \in \{0, 1\}$. Thus, the learner's learning process is represented as a time series $X = \{x_1, x_2, \ldots, x_T\}$, where T represents the last time step [48]. Given a learner's previous practice sequence X, the goal is to model the learner's interaction and thus trace the progression of the learner's knowledge proficiency over time [49]. Based on this, we aim to predict the probability $p(a_{t+1} = 1|X, q_{t+1})$ of the learner correctly answering a question q_{t+1}. For convenience, Table 1 contains a list of symbols used in this paper.

Definition 1 (*Multi-association Heterogeneous Graph*)**.** In this paper, the multi-association heterogeneous graph is represented as $G(V, R)$; entities are denoted by $V \in \{Q, C\}$, where $Q \in \{q_1, q_2, \ldots, q_m\}$ denotes the question set and $C \in \{c_1, c_2, \ldots, c_n\}$ the concept set; r denotes a specific type of association (correlation, prerequisite between questions, similarity between questions, etc.) in $R \in \{r_1, r_2, \ldots, r_k\}$, and k is the total number of association types.

Definition 2 (*Knowledge Triplet*)**.** The association between questions and concepts can be transformed into a knowledge triplet (h, r, t), which is derived from the set $S = \{(h, r, t)\,|(h, r, t) \in G\}$, for $h, t \in \{Q, C\}$ and $r \in R$.

Definition 3 (*Association Weight Matrix*)**.** For each association r, a weight matrix $A^r \in \mathbb{R}^{(n+m)\times(n+m)}$, where $0 < A^r_{i,j} < 1$ represents the weight value of the association between entities i and j.

4. The proposed MAHKT

As illustrated in Fig. 2, we present an overview flowchart of the MAHKT method. Specifically, when predicting the learner's future performance, the multi-association heterogeneous graph embedding module first constructs the heterogeneous graph by comprehensively considering five potential associations. After extracting knowledge triples from the graph, MAHKT further adopts triplet embedding methods to embed these knowledge triples for downstream processing. Then the knowledge propagation through near and far transfer module computes knowledge-aware attention weights to perform near-transfer knowledge propagation for the first-layer neighbor nodes and far-transfer propagation for multi-layer neighbor nodes. Finally, the knowledge state evolution via dual-LSTM and performance prediction module leverages the dual-LSTM to trace the temporal evolution of the learner's knowledge states across the question and concept dimensions. The two dimensions are fused to form the current knowledge state representation, which is used to predict the learner's future responses. The detailed overview of MAHKT is shown in Fig. 3.

4.1. Multi-association heterogeneous graph embedding

To capture implicit knowledge associations between questions and concepts and enrich question representation, we first constructed a heterogeneous graph with five types of associations, converted it into knowledge triplets, and then applied three embedding methods to obtain embedding vectors.

Constructing a multi-association heterogeneous graph. According to Definition 1, a multi-association heterogeneous graph includes two types of entities (questions and concepts) and various edges (five types of associations), as shown in the construction process in Fig. 3(a). First, a conceptual co-correctness matrix U^c is constructed as follows:

$$U^c_{ij} = \begin{cases} u^c_{ij}, & i \neq j \\ 0, & i = j \end{cases}, \tag{1}$$

where u^c_{ij} represents the number of times concept j is answered correctly following a correct response to concept i. Next, using the conceptual co-correctness matrix, three methods are applied to calculate the five types of edge associations defined in Fig. 1(b):

(1) Prerequisite Associations. In addition to the common prerequisite associations among concepts found in existing models, our approach also includes prerequisites among questions [12]. The calculation methods for these two types of associations are similar. Therefore, we use concept prerequisites as an example. We constructed a concept prerequisite matrix T^c, where $T^c_{i,j} \in \{0, 1\}$ indicates whether concept i is a prerequisite for concept j, as follows:

$$\tilde{T}^c_{ij} = \frac{u^c_{ij}}{\sum_k u^c_{ik}}, \tag{2}$$

where $i \neq j$; otherwise, $\tilde{T}^c_{ij}$ is set to 0. Then, if $\tilde{T}^c_{ij} > \theta$, $T^c_{ij} = 1$; otherwise $T^c_{ij} = 0$, and θ is set to the average value of $\tilde{T}^c$. We removed the loops to maintain a directed acyclic graph.

(2) Similarity Associations. In addition to the common similarity associations among questions found in existing models, our approach includes similarity associations among concepts. Using concepts as examples, we constructed a concept similarity matrix O^c, where $O^c_{i,j} \in \{0, 1\}$ indicates whether concept i is similar to concept j:

$$\tilde{O}^c_{ij} = \frac{u^c_{ij} + u^c_{ji}}{\left|u^c_{ij} - u^c_{ji}\right| + \varepsilon}, \tag{3}$$

where $\varepsilon = 0.1$ is used to prevent division by zero. Subsequently, the maximum and minimum scale methods were utilized to obtain the normalized similarity association coefficients $\tilde{O}^c$:

$$\tilde{O}^c_{ij} = \frac{\tilde{O}^c_{ij} - \min\left(\tilde{O}^c\right)}{\max\left(\tilde{O}^c\right) - \min\left(\tilde{O}^c\right)}, \tag{4}$$

Finally, the concept similarity matrix O^c was obtained based on threshold θ. Specifically, if $\tilde{O}^c_{ij} > \theta$, then $O^c_{ij} = 1$; otherwise $O^c_{ij} = 0$,

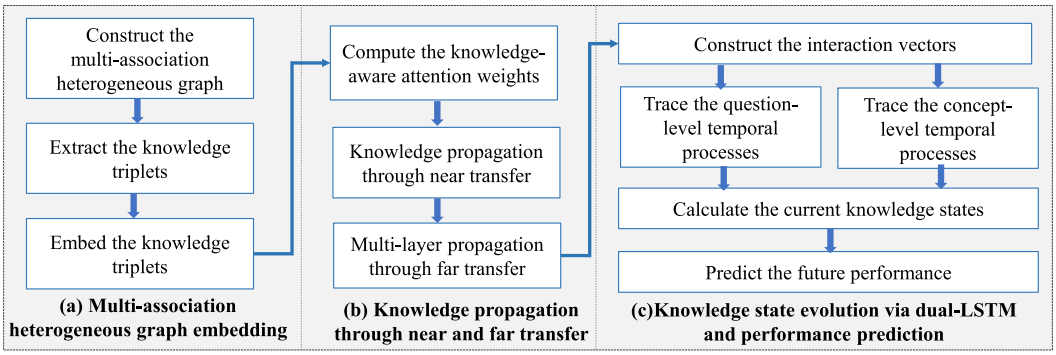


Fig. 2. The overview flowchart of the MAHKT method consists of three modules: (a) constructing a multi-association heterogeneous graph with five associations and extracting knowledge triples; (b) performing near- and far-transfer using knowledge-aware attention, aggregating neighboring knowledge, and propagating multi-layer representations to capture higher-order dependencies; and (c) evolving knowledge states with a dual-LSTM to trace learners' knowledge across questions and concepts, predicting future responses.

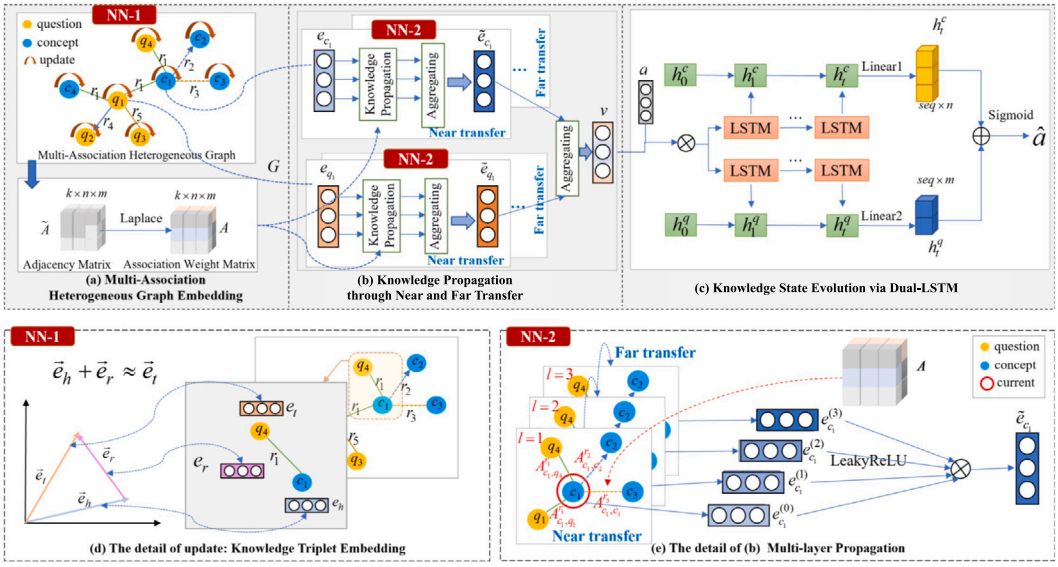


Fig. 3. The detailed overview of MAHKT framework. MAHKT consists of three primary modules: (a) Multi-Association Heterogeneous Graph Embedding, (b) Knowledge Propagation through Near and Far Transfer, and (c) Knowledge State Evolution via Dual-LSTM and Performance Prediction. The specific processes for the first two modules are detailed as NN-1 and NN-2, shown in parts (d) and (e).

setting θ to the average value of $\tilde{O}^c$. The motivation for this setting is to ensure that the model is stable and generalizable when dealing with different types of data distributions.

(3) Correlation Association. The correlation is depicted through a commonly used Q matrix, where each $Q_{i,j} \in \{0,1\}$ represents whether question i is correlated with concept j.

Extracting knowledge triplets. In the multi-association heterogeneous graph, each edge can be converted into a knowledge triple (h,r,t), e.g., the association $c_1 \xrightarrow{r_1} q_4$ in Fig. 3(d) can be transformed into a knowledge triplet (c_1, r_1, q_4), which means that there is an association r_1 between concept c_1 and question q_4. Each association r is represented by a weight matrix A^r, which is a 2D matrix with rows and columns corresponding to questions or concepts, and the values represent correlation strength. Specifically, we constructed an adjacency matrix $\tilde{A}^r$:

$$\tilde{A}^r_{i,j} = \begin{cases} 1, & \text{if there exists } r \text{ between } i \text{ and } j \\ 0, & \text{otherwise} \end{cases}, \tag{5}$$

Furthermore, we constructed a degree matrix D^r, where the elements along the diagonal are the degrees of the individual nodes. The formula is as follows:

$$D^r_{i,j} = \begin{cases} \sum_{j=1}^{n+m} \tilde{A}^r_{i,j}, & \text{if } j = i \\ 0, & \text{otherwise} \end{cases}, \tag{6}$$

Finally, based on identity matrix I, adjacency matrix $\tilde{A}^r$, and degree matrix D^r, we calculated the association weight matrix A^r using the random walk normalized Laplacian algorithm:

$$A^r = I - D^{r^{-1}} \tilde{A}^r, \tag{7}$$

Knowledge triplet embedding. Knowledge triple embedding is an efficient approach for representing questions and concepts as embedding vectors, while preserving the underlying graph structure [50]. In addition, it offers good scalability, allowing the expansion of existing associations to enhance question and concept embedding. Three approaches were employed to implement knowledge triple embeddings. The complete process is presented in algorithm 1.

For each knowledge triplet (h,r,t), the embedding vectors for each question and concept are learned by optimizing the transformation principle $e^r_h + e_r \approx e^r_t$ [51], as shown in Fig. 3(d). Here, e_h, e_t, and $e_r \in \mathbb{R}^d$ are the embeddings of h, t, and r, respectively; e^r_h and e^r_t are projections of e_h and e_t in the associated space. Hence, we implemented a likelihood score formula using three methods:

Algorithm 1: Knowledge Triplet Embedding

Input: Knowledge triplets set S, questions, concepts and associations set Q, C and R, embeddings dimensions d.
Output: Embedding e_h, e_t of questions and concepts, respectively.

Initialize: $e_h \leftarrow xavier\ for\ each\ q \in Q, c \in C$,
$e_t \leftarrow xavier\ for\ each\ q \in Q, c \in C$,
$e_r, w_r, M_r \leftarrow xavier\ for\ each\ r \in R$.
while *epoch and not converged* **do**
  $e \leftarrow e/\|e\|\ \ for\ each\ q \in Q, c \in C$
  $S_{batch} \leftarrow sample(S, b)$ // sample a batch of size b
  $T_{batch} \leftarrow \phi$ // initialize the set of knowledge triplets
  for $(h, r, t) \in S_{batch}$ **do**
    $(h, r, t') \leftarrow sample(S'_{(h,r,t)})$ // sample a corrupted knowledge triplet $T_{batch} \leftarrow T_{batch} \bigcup \{(h, r, t), (h, r, t')\}$
  end
  if *TransE* **then**
    calculate $g(h, r, t)$ by Equation 8 ;
  else
    if *TransH* **then**
      update normal vector w_r ;
      calculate $e_{h_\perp}$ via $e_h - w_r^T e_h$;
      calculate $e_{t_\perp}$ via $e_h - w_r^T e_t$;
      $g(h, r, t) \leftarrow -\left\| e_{h_\perp} + e_r - e_{t_\perp} \right\|_2^2$;
    else
      update projection vector M_r ;
      calculate $e_{h_\perp}$ with M_r and e_h ;
      calculate $e_{t_\perp}$ with M_r and e_t ;
      $g(h, r, t) \leftarrow -\left\| e_{h_\perp} + e_r - e_{t_\perp} \right\|_2^2$;
    end
  end
  calculate loss $\mathcal{L}$ by Equation 11
end

- TransE: Questions and concepts are represented as vectors e_h and e_t. The likelihood score is shown in the following formula:

$$g_{TransE}(h, r, t) = -\left\| e_h + e_r - e_t \right\|_{1/2}, \tag{8}$$

- TransH: Questions and concepts are represented as vectors in a hyperplane with a normal vector w_r. By projecting h and t onto the hyperplane, denoted as $e_{h_\perp} = e_h - w_r^T e_h w_r$, $e_{t_\perp} = e_t - w_r^T e_t w_r$. The likelihood score is shown in the following formula:

$$g_{TransH}(h, r, t) = -\left\| e_{h_\perp} + e_r - e_{t_\perp} \right\|_2^2, \tag{9}$$

- TransR: Questions and concepts are represented as vectors in a solid space with h and t projected onto this space, i.e., M_r, $e_{h_\perp} = M_r e_h, e_{t_\perp} = M_r e_t$. The likelihood score is shown in the following formula:

$$g_{TransR}(h, r, t) = -\left\| e_{h_\perp} + e_r - e_{t_\perp} \right\|_2^2, \tag{10}$$

To capture the differences between the pseudo- and true-knowledge triplets, we designed the loss function of the above three embedding methods as follows:

$$\mathcal{L} = \sum_{(h,r,t,t')} -\ln \sigma\left(g(h, r, t') - g(h, r, t)\right), \tag{11}$$

where $(h, r, t') \notin G$ is the constructed pseudo-knowledge triplet derived by randomly replacing an entity in a real knowledge triplet, and $\sigma(\cdot)$ denotes the sigmoid activation function. Finally, we obtain embedding vectors e_h and e_t of questions and concepts, respectively.

4.2. Knowledge propagation through near and far transfer

To effectively extract and leverage the rich underlying information among associations, we propose two knowledge propagation patterns that follow near- and far-transfer theories: near-transfer-based neighbor aggregation and far-transfer-based recursive embedding. To this end, we employed an attention mechanism to implement the propagation of node information according to node similarity weights, as illustrated in Fig. 3(b).

Knowledge-aware attention. According to the theory of common elements [16], the likelihood of knowledge transfer increases with the number of features shared between entities. Therefore, the association weight matrix is first computed using an attention mechanism to update the weight matrix of each association, thereby highlighting the similarity between nodes. Specifically, the update method for the association weight matrix A^r is as follows:

$$A_{h,t}^r = e_t{}^{\mathrm{T}} \tanh\left(e_h + e_r\right), \tag{12}$$

where the tanh activation function makes the attention score depend on the distance between e_h and e_t in the associated r space; for example, when entities are closer to each other, they propagate more information through higher attention scores in near transfer. Subsequently, the softmax function is applied to normalize the coefficients of all knowledge triplets concatenated to h:

$$A_{h,t}^r = \frac{\exp\left(A_{h,t}^r\right)}{\sum_{(h,r,t') \in N_h} \exp\left(A_{h,t'}^r\right)}, \tag{13}$$

Therefore, the final attention scores indicate the importance of different neighboring entities in capturing shared features, thereby determining whether near or far transfer patterns will be adopted during the node aggregation process. In other words, during forward propagation, attention scores directly affect the perceived scope and degree of neighboring node information.

Knowledge propagation through near transfer. To implement near-transfer-based knowledge propagation, the influence of neighboring nodes on the current node is regulated based on knowledge-aware attention, which enables neighborhood information transfer through associations. Specifically, the knowledge triplet is treated as the basic propagation unit, where the question or concept is a bridge for connecting the propagation units and delivering the information. For example, $c_2 \xrightarrow{-r_2} c_1 \xrightarrow{r_1} q_1$ in Fig. 3, the concept c_1 takes concept c_2 and question q_1 as inputs to enrich its own features. Note that nodes with more shared features receive higher attention scores, making them more likely to undergo near transfer. Considering a head entity h, its first-layer connectivity structure, e_{N_h}, is captured as follows:

$$e_{N_h} = \sum_{(h,r,t) \in N_h} A_{h,t}^r e_t, \tag{14}$$

where $A_{h,t}^r$ controls the weight of propagation for each knowledge triple, indicating the importance of knowledge propagation from t to h under association r. Subsequently, the embedding vector of entity h is updated based on the initial embedding vector e_h and neighboring node aggregation vector e_{N_h}, $e_h^{(1)} = f\left(e_h, e_{N_h}\right)$. More specifically, the aggregator f is implemented as follows:

$$f\left(e_h, e_{N_h}\right) = \mathrm{LeakyReLU}(W_1(e_h + e_{N_h})) + \mathrm{LeakyReLU}(W_2(e_h \odot e_{N_h})), \tag{15}$$

where $W_1, W_2 \in \mathbb{R}^{d' \times d}$ are the trainable weight matrices, and $\odot$ denotes the element-wise product. After these calculations and near transfer-based propagation, the embedding vectors $e^{(1)}$ of the questions and concepts that contain the first-layer knowledge information are obtained.

Multi-layer propagation through far transfer. Although far transfers are less common than near transfers [18], several scholars have

validated their existence [15,20]. For example, $c_1 \xrightarrow{r_1} q_1 \xrightarrow{r_4} q_2$ in Fig. 3, besides the existence of near transfer effects between c_1 and q_1, there is an indirect far near effect between q_2 and c_1. In this section, we explore the information from indirectly connected nodes by stacking propagation layers and capturing the far-transfer effects from multilayer neighbors, as shown in Fig. 3(e). A question or concept in layer l is recursively represented as follows:

$$e_h^{(l)} = f\left(e_h^{(l-1)}, e_{N_h}^{(l-1)}\right), \quad e_{N_h}^{(l-1)} = \sum_{(h,r,t)\in N_h} A_{h,t}^r e_t^{(l-1)}, \tag{16}$$

where $e_t^{(l-1)}$ represents the embedding of entity t generated by the previous knowledge propagation layer, preserving information from $(l-1)$-layer neighbors. During the first iteration of information propagation, $e_h^{(0)}$ is initialized as e_h.

Therefore, by utilizing the following three modules: knowledge-aware attention, knowledge propagation through near transfer, and multilayer propagation through far transfer, the multilayer connected node information can be captured based on knowledge-aware attention scores, to obtain both near and far transfer effects. For example, in $c_2 \xrightarrow{-r_2} c_1 \xrightarrow{r_1} q_1 \xrightarrow{r_4} q_2$, question q_2 can aggregate information from all connected nodes, and the comprehensive representation of question q_2 is explicitly embedded in $e_{c_2}^{(3)}$. After obtaining the enhanced question embedding $\tilde{e}^q$ and concept embedding $\tilde{e}^c$, by utilizing the aggregator f mentioned above to aggregate $\tilde{e}^q$ and $\tilde{e}^c$, fusion knowledge embedding $v \in \mathbb{R}^{2d}$ can be generated.

4.3. Knowledge state evolution via dual-LSTM architecture

To effectively trace learner's knowledge states, MAHKT utilizes a dual-long short-term-memory (LSTM) architecture that models the learning process not only along the concept dimension but also along the question dimension. These modules comprehensively capture both near- and far-transfer effects during knowledge propagation, thereby mitigating the incomplete tracking caused by data sparsity.

Constructing interaction vectors. To differentiate between correct and incorrect answers, we performed element-wise multiplication between the fused knowledge embedding v and response a, concatenating the results to obtain the interaction vector x.

$$x = (v \otimes a) \oplus (v \otimes (1-a)), \tag{17}$$

where $\oplus$ denotes the concatenated operations; $x \in \mathbb{R}^{2d}$ denotes an interaction vector.

Tracing temporal processes. Because the KT problem can be regarded as a prediction problem based on long-term time-series data, we employed LSTM to trace the learner's knowledge state and extract relevant information from the past learning history [43].

Insights from human teachers have revealed that questions and concepts are distinct entities. Studies have shown that tracking concepts and questions separately leads to a more accurate knowledge state assessment. Therefore, to more precisely estimate the knowledge state, we implemented a dual-LSTM architecture by conducting temporal process tracing separately along the dimensions of concepts and questions. In step t, the learner's hidden knowledge state h_t is updated based on the current input and previous state.

$$h_t = \mathrm{LSTM}\left(x_t, h_{t-1}\right), \tag{18}$$

The dual-LSTM neural network outputs h_t^q and h_t^c are then passed through a linear layer to perform dimension conversion, yielding the question- and concept-level knowledge states, as follows:

$$\begin{aligned} h_t^q &= W_1 h_t^q + b_1, \\ h_t^c &= W_2 h_t^c + b_2, \end{aligned} \tag{19}$$

where W_1 and W_2 are the weight parameters, and b_1 and b_2 are the bias parameters.

Calculating current knowledge states. The knowledge states from question h_t^q and concept h_t^c were averaged to obtain the comprehensive state $\bar{h}_t$ at the current time step t using the following formula:

$$\bar{h}_t = \left(h_t^q + h_t^c\right)/2, \tag{20}$$

Table 2
The statistics of datasets.

Dataset	ASSIST2017	JUNYI2015	NIPS34	PEIYOU
# Records	942,816	25,925,984	1,382,727	5,549,635
# Learners	686	191,874	4,918	85,600
# Questions	3162	721	948	7652
# Concepts	102	39	57	865

4.4. Final performance prediction

Because of the unobservable nature of knowledge states, KT models typically evaluate performance by predicting future outcomes [23,30]. Accordingly, our MAHKT model employs a similar approach, outputting the probability that a learner will correctly answer a new question.

Future performance prediction. First, the hidden knowledge state from the current time step $\bar{h}_t$ is concatenated with the embedding of the next question v_{t+1} and input into MAHKT. A sigmoid function is then applied to generate the predicted probability $\hat{a}_{t+1} \in [0,1]$, where a higher $\hat{a}_{t+1}$ value indicates a greater likelihood of a correct response. The prediction method is calculated as follows:

$$\hat{a}_{t+1} = \mathrm{Sigmoid}\left(W_{\mathrm{pre}}\left(\bar{h}_t \oplus v_{t+1}\right) + b_{\mathrm{pre}}\right), \tag{21}$$

Objective function. To optimize our model, we used binary cross-entropy to calculate the loss between the predicted value $\hat{a}$ and the true answer a. The objective function is defined as

$$\mathcal{L}(\hat{a}_t, a_t) = -\sum_{t=1}^{T}\left(a_t \log \hat{a}_t + \left(1-a_t\right)\log(1-\hat{a}_t)\right). \tag{22}$$

5. Experiments

In this section, we describe the comprehensive experiments conducted on four real-world datasets to explore the following research questions, which validate the contributions of our study:

RQ1: How does MAHKT compare to state-of-the-art knowledge tracing methods?

RQ2: What is the impact of the knowledge embedding and propagation modules on MAHKT's performance?

RQ3: How do different knowledge state learning approaches influence MAHKT's effectiveness?

RQ4: How do various knowledge triple embedding methods affect the performance of MAHKT?

5.1. Experimental setup

5.1.1. Datasets

To evaluate the effectiveness of MAHKT, we performed extensive experiments on four publicly available benchmark datasets. The statistics for these datasets are presented in Table 2.

ASSIST2017[1]: This dataset consists of math problem-solving records collected from middle and high school learners through the ASSISTments tutoring system.

JUNYI2015[2]: This dataset comprises performance logs of mathematical exercises by learners and a knowledge structure labeled by experts.

[1] https://sites.google.com/view/assistmentsdatamining/dataset?authuser=0.

[2] https://www.junyiacademy.org/.

Table 3
Performance comparison on the KT task.

Model		DKT	DKVMN	DKT+	DIRT	GKT	AKT	ATKT	GIKT	DKT-Q	IEKT	ATDKT	MAHKT
ASSIST 2017	AUC	0.7095	0.7061	0.7210	0.7091	0.7224	0.7459	0.7292	0.7507	0.7662	0.7561	0.7368	**0.7943**
	ACC	0.6848	0.6813	0.6889	0.6928	0.6913	0.7018	0.6928	0.7055	0.7151	0.7087	0.6955	**0.7352**
	RMSE	0.4510	0.4518	0.4473	0.4509	0.4469	0.4399	0.4447	0.4362	0.4308	0.4346	0.4436	**0.4187**
	MAE	0.4017	0.4069	0.3978	0.4069	0.4019	0.3752	0.3942	0.3782	0.3688	0.3772	0.3797	**0.3448**
JUNYI 2015	AUC	0.7234	0.7372	0.7381	0.7353	0.7383	0.7836	0.7371	0.7706	0.7851	0.7911	0.7673	**0.7927**
	ACC	0.8395	0.8422	0.8420	0.8420	0.8420	0.8477	0.8418	0.8432	0.8475	0.8472	0.7924	**0.8503**
	RMSE	0.3528	0.3491	0.3491	0.3495	0.3491	0.3390	0.3494	0.3439	0.3394	0.3360	0.3798	**0.3358**
	MAE	0.2500	0.2446	0.2433	0.2448	0.2466	0.2282	0.2438	0.2375	0.2307	0.2253	0.2889	**0.2244**
NIPS34	AUC	0.7659	0.7646	0.7721	0.7692	0.7696	0.7899	0.7761	0.7920	0.7995	0.8008	0.7859	**0.8064**
	ACC	0.7005	0.6984	0.7055	0.7025	0.7027	0.7136	0.7076	0.7217	0.7285	0.7282	0.7176	**0.7346**
	RMSE	0.4413	0.4419	0.4383	0.4398	0.4397	0.4257	0.4365	0.4290	0.4246	0.4245	0.4319	**0.4209**
	MAE	0.3886	0.3914	0.3819	0.3863	0.3878	0.3611	0.3805	0.3695	0.3579	0.3592	0.3740	**0.3542**
PEIYOU	AUC	0.8160	0.8156	0.8223	0.8153	0.8043	0.8383	0.7936	0.8110	0.8485	0.8510	0.8431	**0.8524**
	ACC	0.8249	0.8241	0.8270	0.8236	0.8309	0.8320	0.7788	0.8374	0.8374	0.8369	0.8345	**0.8386**
	RMSE	0.3546	0.3551	0.3522	0.3558	0.3549	0.3462	0.3845	0.3534	0.3406	0.3410	0.3435	**0.3387**
	MAE	0.2479	0.2491	0.2427	0.2515	0.2543	0.2387	0.2816	0.2537	0.2285	0.2303	0.2418	**0.2281**

NIPS34[3]: This dataset is derived from tasks 3 and 4 of the Neural Information Processing Systems (NeurIPS) 2020 Education Challenge. It contains learner answers to multiple-choice diagnostic math problems collected from the Equity, Equality, Diversity & Inclusion (EEDI) platform.

PEIYOU[4]: This dataset is a public resource from the Global Knowledge Tracing Challenge @AAAI2023, created by a research team at Beijing Normal University and contains answer records from fourth-grade learners in China.

5.1.2. Baselines

The proposed model was compared with eleven state-of-the-art baseline models, which were categorized into two groups based on whether they traced learners' knowledge states at the concept or question level.

- **DKT** [10] updates a learner's knowledge state by feeding concept representations into an RNN and using the resulting state to predict the learner's answers.
- **DKVMN** [52] employs a key–value memory network to store and update the learner's knowledge state for all concepts during the interaction with a question.
- **DKT+** [53] is an extended variant of DKT, designed to address input issues that cannot be reconstructed and observed in DKT.
- **DIRT** [54] uses proficiency vectors to represent learners' understanding of knowledge concepts and applies probabilistic response functions to predict their performance.
- **GKT** [33] is KT's first method to use the GNN, which represents conceptual associations as graphs and formulates the KT task as a time-series node-level classification task.
- **AKT** [55] is an attention-based KT model that builds context-aware question representations and uses a novel monotonic attention mechanism to connect a learner's current answers with their past responses.
- **ATKT** [27] applies adversarial training techniques, such as adversarial perturbation, to raw learner interaction sequences to mitigate DKT's risks of overfitting and limited generalization.
- **GIKT** [34] is a GCN-based KT method, which aims to address the problem that the association between questions and concepts is not fully captured by information aggregation and propagation.
- **DKT-Q** is a variant of DKT that replaces concept embeddings with one-shot encodings of questions as input.
- **IEKT** [56] is a single-state RNN-based model. It incorporates the learner's characteristics into the model by considering the learner's cognitive level and knowledge acquisition sensitivity to the problem.
- **ATDKT** [57] achieves effective prediction of learners' future performance by designing two novel auxiliary tasks through the multitask framework.

5.1.3. Experimental setup

Knowledge propagation through the near- and far-transfer modules consisted of three layers with hidden dimensions of 64, 32, and 16. Knowledge state evolution via the Dual-LSTM architecture module had a hidden layer size of 200 and a dropout rate of 0.1 [58]. Hyperparameters were determined using a grid search, and all parameter matrices were initialized and updated using Xavier initialization during training [59]. We used 128 dimensions for concept and question embedding in the knowledge triple-embedding module. For all four datasets, we employed the SGD optimizer with a learning rate of 0.01 for the knowledge triplet embedding module, and adaptive moment estimation (Adam) was used as the optimizer with a learning rate of 0.001 for knowledge propagation through near- and far-transfer modules. The batch size was set to 24, and early stopping was implemented if the validation loss did not decrease within 20 consecutive epochs [60]. To comprehensively evaluate the performance of MAHKT, we utilized area under the receiver operating characteristic curve (AUC), accuracy (ACC), root mean-square error (RMSE), and mean absolute error (MAE) as the experimental evaluation metrics.

5.2. Performance prediction (RQ1)

To evaluate the predictive effectiveness of the MAHKT model, we compared it with state-of-the-art baseline models by predicting the learners' future responses. The results for the four datasets, shown in Table 3, provide several key insights. First, MAHKT consistently outperformed all other models, with a particularly notable improvement on the ASSIST2017 dataset, where it achieved a 2.81% higher AUC score than the best-performing baseline. This underscores the superior predictive accuracy of this model. Second, question-level models generally exhibit better performance than concept-level models, possibly because of the limited number of concepts in these datasets, whereby each concept is mapped to multiple questions, thus complicating accurate mastery prediction [29]. In addition, among concept-level models, AKT demonstrated a marked advantage over other RNN-based models, highlighting the efficacy of attention mechanisms [61]. The graph-based knowledge tracing method, represented by GIKT, outperforms models such as AKT and ATKT on the two datasets, demonstrating that capturing the relationships between entities through graph-based methods can

[3] https://eedi.com/projects/neurips-education-challenge.
[4] https://ai4ed.cc/competitions/aaai2023competition.

Table 4
Parameter sensitivity experiments for the number of LSTM layers n_l and the embedding dimension d of MAHKT. We highlight the best results with bold.

Parameter	ASSIST2017			JUNYI2015			NIPS34			PEIYOU		
	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
MAHKT-$n_l = 1$	**0.7943**	**0.7352**	**0.4187**	**0.7927**	**0.8503**	**0.3358**	0.8064	0.7346	0.4209	**0.8524**	**0.8386**	**0.3387**
MAHKT-$n_l = 2$	0.7925	0.7343	0.4192	0.7916	0.8500	0.3361	0.8070	**0.7356**	0.4203	0.8523	0.8384	0.3388
MAHKT-$n_l = 3$	0.7930	0.7331	0.4191	0.7898	0.8497	0.3366	**0.8071**	**0.7356**	**0.4202**	0.8521	**0.8386**	0.3388
MAHKT-$d = 32$	0.7935	0.7344	0.4190	0.7918	0.8501	0.3360	0.8059	0.7339	0.4212	0.8519	0.8385	0.3389
MAHKT-$d = 64$	0.7930	0.7343	0.4191	0.7925	0.8501	0.3358	0.8063	**0.7346**	**0.4209**	0.8521	0.8385	0.3388
MAHKT-$d = 128$	**0.7943**	**0.7352**	**0.4187**	0.7927	**0.8503**	0.3358	**0.8064**	**0.7346**	**0.4209**	**0.8524**	**0.8386**	**0.3387**
MAHKT-$d = 256$	**0.7943**	0.7351	0.4189	**0.7929**	0.8502	**0.3357**	0.8062	0.7340	0.4210	**0.8524**	0.8385	0.3388

Table 5
Parameter comparison experiments for the value of θ. We highlight the best results with bold.

Parameter θ	ASSIST2017			JUNYI2015			NIPS34			PEIYOU		
	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
$\theta = 0.00001$	0.7937	0.7344	0.4190	0.7927	0.8502	0.3358	0.8059	0.7343	0.4210	0.8521	0.8384	0.3390
$\theta = 0.25$	0.7942	**0.7356**	**0.4186**	0.7927	0.8503	**0.3357**	0.8063	0.7346	0.4210	0.8523	0.8382	0.3389
$\theta = 0.50$	0.7941	0.7354	0.4187	**0.7929**	0.8503	**0.3357**	0.8062	**0.7348**	0.4209	**0.8524**	0.8384	0.3388
$\theta = 0.75$	0.7939	0.7353	0.4187	0.7928	**0.8504**	**0.3357**	**0.8064**	0.7345	**0.4208**	0.8523	**0.8386**	0.3389
$\theta = 0.99999$	0.7936	0.7347	0.4189	0.7928	0.8502	0.3358	0.8060	0.7342	0.4209	0.8522	0.8385	**0.3387**
θ−avg	**0.7943**	0.7352	0.4187	0.7927	0.8503	0.3358	**0.8064**	0.7346	0.4209	**0.8524**	**0.8386**	**0.3387**

enhance question representations and improve model performance to some extent. However, GIKT still performs poorly compared to MAHKT, despite utilizing the similar propagation mechanisms. This difference may be due to the fact that GIKT lacks sufficient modeling of the five associations between questions and concepts, especially the prerequisite associations. Furthermore, while GIKT uses GCN to aggregate and propagate information, MAHKT flexibly adjusts knowledge propagation paths through knowledge-aware attention, capturing both short- and long-range knowledge associations, aligning better with learners' cognitive development laws and promoting effective knowledge transfer. These differences may contribute to MAHKT's ability to model complex relationships and achieve superior performance.

To evaluate the impact of hyperparameters such as the number of LSTM layers n_l and embedding dimension d on the MAHKT, we provide detailed parameter sensitivity experiments on the number of LSTM layers and the embedding dimension, as shown in Table 4. According to the experimental results, we find that when the number of LSTM layers n_l is set to 1, the model's AUC metric performs best on all three datasets. In addition, we also notice that when the embedding dimension d is 128 or 256, the AUC of the MAHKT can be optimal on all three datasets. However, the model's ACC and RMSE are suboptimal on multiple datasets when $d = 256$. Therefore, in order to balance the model's performance and computational cost, we finally set the number of LSTM layer n_l and embedding dimension d to 1 and 128, respectively. In addition, MAHKT can consistently outperform other baseline models across all tested hyperparameter ranges. These experimental results demonstrate the effectiveness and robustness of MAHKT.

To further explore the impact of θ on the performance of MAHKT, we conduct the parameter comparison experiments for the value of θ, as presented in Table 5. "θ−avg" denotes the θ is set to the average value. Based on the experimental results, we find that MAHKT achieves the best performance on the three datasets when θ is set to the average value. Although some fixed θ values are slightly better than the "θ−avg" on some metrics, "θ-avg" still provides stable and balanced performance for MAHKT. This indicates that θ-avg is able to effectively strike the optimal balance between stability and generalization ability in different learning scenarios, which further demonstrates the validity of our motivation for setting θ to the average value.

5.3. Ablation analysis of knowledge embedding and propagation (RQ2)

To further evaluate the impact of the knowledge embedding and propagation module on prediction performance, we conducted ablation experiments to assess the contributions of different components within MAHKT. As shown in Table 6, the knowledge propagation module comprises three key components. We denote the exclusion of any of these components from MAHKT with a "-" sign. Specifically, MAHKT-none refers to a model variant in which all components are removed. MAHKT-E omits the knowledge triplet embedding module, which means that the association information between questions and concepts is not utilized for embedding updates. MAHKT-N&F excludes near- and far-transfer effects; therefore, the model does not aggregate information from multilevel neighboring entities. Finally, MAHKT-F eliminates far transfer in knowledge propagation by reducing the number of propagation layers to one, thereby incorporating only near transfer (in contrast, the proposed MAHKT model uses three layers). Notably, evaluating a scenario in which far transfer occurs without near transfer is impossible because near transfer inherently occurs whenever knowledge propagation is present [17]. Detailed model descriptions and results are provided in Table 6.

First, removal of any module from MAHKT resulted in a significant decline in performance, underscoring the critical role of each module in achieving accurate predictions. As expected, the complete omission of both heterogeneous graph embeddings and knowledge propagation modules led to poor performance. Additionally, MAHKT-E exhibited a more pronounced drop in AUC than MAHKT-N&F, indicating that the heterogeneous graph-embedding module exerts a more substantial impact on the model performance than the knowledge propagation module. This finding suggests that the intrinsic information within entities is more crucial than the information derived from neighboring entities [12].

Furthermore, MAHKT consistently outperformed MAHKT-F&N, demonstrating that the incorporation of both near- and far-transfer effects, which aggregate multilayer entity information, is essential for enhancing question and concept embeddings. This underscores the significance of near- and far-transfers, suggesting that these effects are indeed prevalent [14,15]. Conversely, the similar performances of MAHKT-F&N and MAHKT-F indicate that far transfer has a relatively minor influence on the model, supporting the notion that far transfer exists and is comparatively rare [15,19]. These experiments offer compelling empirical evidence for the design principles behind MAHKT, underscoring the importance of knowledge propagation based on near and far transfer.

5.4. Impact of knowledge state learning (RQ3)

To further evaluate the effectiveness of the proposed dual-LSTM-based knowledge state-learning module, we conducted ablation experiments on four datasets, and the results are presented in Table 7.

Table 6
Settings and AUC metric comparison of MAHKT variants.

	Graph embedding	Near transfer	Far transfer	ASSIST2017	JUNYI2017	NIPS34	PEIYOU
MAHKT-none	×	×	×	0.7774	0.7896	0.8012	0.8377
MAHKT-E	×	✓	✓	0.7808	0.7905	0.8044	0.8487
MAHKT-F&N	✓	×	×	0.7857	0.7905	0.8047	0.8489
MAHKT-F	✓	✓	×	0.7861	0.7914	0.8051	0.8513
MAHKT	✓	✓	✓	**0.7943**	**0.7927**	**0.8064**	**0.8524**

Table 7
Performance comparison of different modeling knowledge state methods.

	Method	AUC	ACC	RMSE	MAE
ASSIST2017	MHAKT-Q	0.7898	0.7316	0.4209	0.3485
	MHAKT-C	0.7807	0.7240	0.4254	0.3541
	MHAKT-Q&C	0.7926	0.7339	0.4195	0.3463
	MHAKT	**0.7943**	**0.7352**	**0.4187**	**0.3448**
JUNYI2015	MHAKT-Q	0.7920	0.8501	0.3360	0.2251
	MHAKT-C	0.7537	0.8434	0.3462	0.2390
	MHAKT-Q&C	0.7919	0.8502	0.336	0.2244
	MHAKT	**0.7927**	**0.8503**	**0.3358**	**0.2241**
NIPS34	MHAKT-Q	0.8059	0.7340	0.4212	**0.3531**
	MHAKT-C	0.7878	0.7191	0.4306	0.3693
	MHAKT-Q&C	0.8064	0.7346	0.4209	0.3542
	MHAKT	**0.8068**	**0.7349**	**0.4206**	0.3534
PEIYOU	MHAKT-Q	0.8510	0.8383	0.3393	**0.2279**
	MHAKT-C	0.8342	0.8318	0.3469	0.2374
	MHAKT-Q&C	0.8505	0.8380	0.3396	0.2280
	MHAKT	**0.8525**	**0.8386**	**0.3387**	0.2281

In the MAHKT-Q variant, only question-level information is utilized, with the knowledge state represented by h_t^q, as specified in Eq. (19). Similarly, MAHKT-C focuses solely on concept-level information, with the knowledge state represented by h_t^c. The MAHKT-Q&C model concatenates question and concept information before feeding into a single LSTM for knowledge-state modeling. By contrast, our full MAHKT model employs two distinct LSTMs to separately capture the learning processes for questions and concepts, subsequently combining them into a comprehensive knowledge state, as described in Eq. (20).

The experimental findings yielded several key observations. First, the MAHKT model consistently outperformed the other models across the AUC, ACC, RMSE, and MAE metrics, suggesting that separately modeling questions and concepts enables the capture of richer information, thereby enhancing the representation of knowledge states. Second, models incorporating both question and concept information (e.g., MAHKT-Q&C and MAHKT) demonstrated superior performance compared with those that rely exclusively on either question or concept data (e.g., MAHKT-Q and MAHKT-C). Even a simple concatenation of question and concept information in MAHKT-Q&C yielded noticeable improvements, indicating that recognizing and leveraging the distinct characteristics of questions and concepts are beneficial. Finally, models focusing on question-level modeling (MAHKT-Q) generally outperformed those centered on concept-level modeling (MAHKT-C), probably because questions contain more detailed information than concepts [30], thereby improving the model's ability to accurately learn and predict the learner's knowledge states. These experiments validate the soundness and effectiveness of our strategy in tracking the evolution of knowledge states using dual LSTM networks.

5.5. Effect of knowledge triple embedding (RQ4)

To investigate the impact of the three multi-association heterogeneous graph embedding methods on the model, we compared the performances of TransE, TransR, and TransH on four datasets across four metrics. The results are presented in Table 8. Performance differences observed among these three knowledge triple embedding methods (TransE, TransR, and TransH) were not significant across the datasets, suggesting that while the embedded information was crucial, the choice of embedding method had a relatively minor impact. This highlights the robustness of the proposed model. Notably, TransH and TransR demonstrated slightly better performance than TransE, possibly because of TransE's limitations in handling 1-N, N-1, and N-N associations [62]. Given that in the KT task, the associations between questions and concepts are categorized as N-N relationships, TransE may struggle to effectively capture this information. Finally, TransR generally provides the best performance, as it leverages the higher-dimensional space to better capture the complexities of N-N relationships, although this comes at the cost of increased computational complexity.

Table 8
Performance comparison of three methods of knowledge triple embedding.

	Method	AUC	ACC	RMSE	MAE
ASSIST2017	TransE	0.7941	0.7347	0.4188	**0.3446**
	TransH	0.7940	0.7348	0.4188	0.3464
	TransR	**0.7943**	**0.7352**	**0.4187**	0.3448
JUNYI2015	TransE	0.7923	0.8501	0.3359	**0.2244**
	TransH	0.7923	0.8502	0.3359	0.2250
	TransR	**0.7927**	**0.8503**	**0.3358**	**0.2244**
NIPS34	TransE	0.8066	0.7340	0.4208	0.3534
	TransH	**0.8068**	**0.7351**	**0.4206**	**0.3522**
	TransR	0.8064	0.7346	0.4209	0.3542
PEIYOU	TransE	0.8523	**0.8387**	**0.3387**	0.2279
	TransH	**0.8524**	0.8386	0.3388	**0.2277**
	TransR	**0.8524**	0.8386	**0.3387**	0.2281

5.6. Effectiveness analysis of graph learning

To investigate the knowledge learned through multi-association heterogeneous graph embedding and knowledge propagation, we conducted visualization analysis on the JUNYI2015 dataset, comparing the clustering effects of the question embeddings generated by our MAHKT model with those produced by the baseline DKT-Q model, as shown in Fig. 4. Specifically, we extracted question embeddings and their initial states, projecting high-dimensional data onto 3D points, to visualize the embedding vectors and assess the clustering effectiveness.

The results indicate that during the initialization phase, both the MAHKT and DKT-Q models exhibited randomly mixed question embeddings. However, after training with heterogeneous graph embedding and knowledge propagation, question embeddings from the MAHKT model demonstrated a clear clustering trend: questions associated with the same concept were clustered together, whereas unrelated questions were distinctly separated. In contrast, the DKT-Q model's embeddings remained randomly distributed even after training and failed to adequately capture the relationships between the questions.

Furthermore, as illustrated in Table 3, although DKT-Q's AUC score is lower than that of the proposed MAHKT model, MAHKT still outperformed the other baseline models possibly because of DKT-Q's use of more comprehensive question embeddings, which allows its LSTM to capture more information and infer the relationships between questions. If DKT-Q could incorporate both near- and far-transfer effects through question and concept embeddings, its performance could potentially be improved. Notably, the DKT-Q model had some outliers, whereas our MAHKT model exhibited only a few, suggesting that DKT-Q may be less stable, further highlighting the robustness of the MAHKT model.

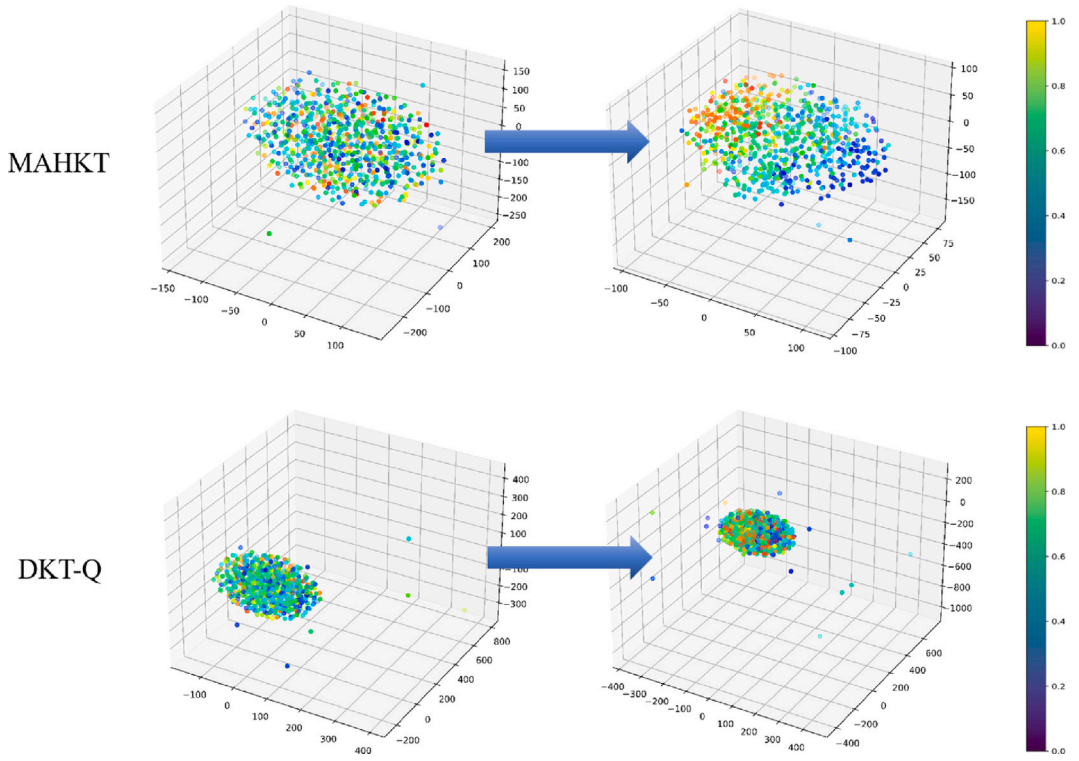


Fig. 4. Visualization of question embeddings in the JUNYI2015 dataset. The top image shows the clustering of the MAHKT model before and after training with heterogeneous graph embedding and knowledge propagation, whereas the bottom image shows the clustering for the DKT-Q model. The MAHKT model achieves better clustering.

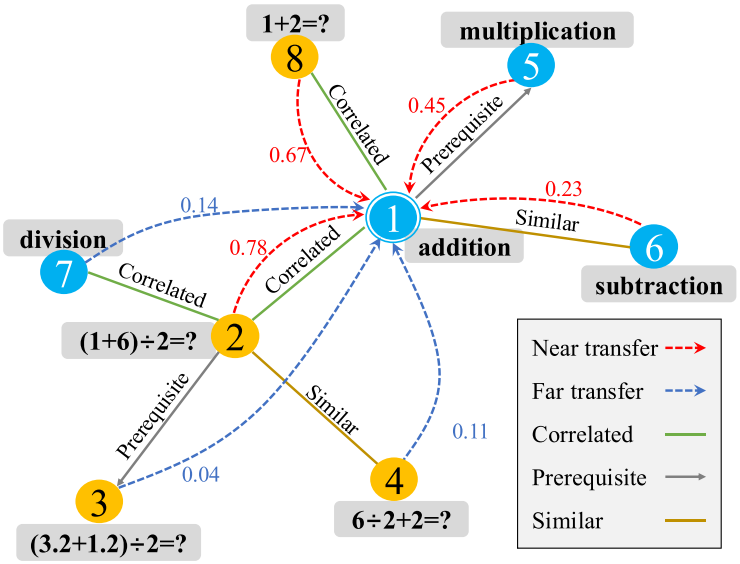


Fig. 5. Visualization of near and far transfer effects cases between concepts and questions in the JUNYI2015 dataset.

5.7. Case analysis

To explore the real associations and interpretability of multi-association heterogeneous graphs after knowledge propagation, we visualize the near- and far-transfer attention weights for four concepts and four questions, randomly selected from the JUNYI2015 dataset, as illustrated in Fig. 5. In the figure, the blue nodes represent concepts and the yellow nodes represent questions. These entities are connected by five distinct types of associations that accurately reflect real-world relationships in elementary mathematics. The figure demonstrates that attention scores for near transfer vary across various associations. For instance, *Concept 1 (Addition)* serves as a prerequisite for *Concept 5 (Multiplication)* with a near-transfer attention score of 0.45, while its similarity-based relationship with *Concept 6 (Subtraction)* is relatively weaker, with a score of 0.23. Such variations illustrate how the model prioritizes information flow during knowledge propagation. Beyond near transfer, far-transfer effects also have the impacts on knowledge propagation, such as the association between *Question 4: 6 ÷ 2 + 2 = ?* and *Concept 1 (Addition)*, with an attention score of 0.11. These far-transfer associations underscore the model's capability to capture long-range dependencies and integrate indirect but still meaningful relationships. Therefore, capturing both near-transfer and far-transfer effects is crucial for enriching entity representations. Furthermore, we

observe that near-transfer attention scores are generally higher than far-transfer scores, supporting the idea that the greater the similarity between entities, the more likely near transfer occurs [18], resulting in higher attention scores. Conversely, when the similarity is lower, far transfer tends to occur [15], but with lower attention scores. By dynamically allocating more attention to the more significant associations, the proposed model effectively propagates knowledge through the graph, enhancing its interpretability and offering deeper insights into the underlying mechanisms of knowledge transfer.

6. Conclusion

In this study, guided by the principles of knowledge transfer theory, we explored the evolution of knowledge states to predict future performance. However, traditional graph-based KT methods overlook the intricate multi-association relationships among entities, leading to deviations in the overall semantic characteristics of the problem representation and inappropriate knowledge transfer propagation. This further results in discrepancies between the actual and represented knowledge states during the transfer process, ultimately affecting the prediction performance of the model. To overcome these limitations, we proposed a novel method called MAHKT, which was designed to enhance both interpretability and performance. MAHKT effectively captures the complex and diverse associations between questions and concepts through a multi-association heterogeneous graph, encompassing prerequisites, similarities, and correlations. The model employs an attention mechanism to quantify node similarity and leverages both near and far transfer mechanisms within knowledge propagation to extract rich associative information. Additionally, MAHKT utilizes dual LSTM models to independently track the evolution of knowledge states for both questions and concepts, providing a more accurate and interpretable representation of learners' evolving knowledge states. Extensive experiments conducted on real-world datasets demonstrate that MAHKT consistently outperforms existing models. Furthermore, ablation studies have confirmed the significant impact of each module on the overall model performance.

Even though our MAHKT model has demonstrated strong performance, several promising directions for future research remain. First, learner performance is not solely determined by correct or incorrect responses; behavioral factors during the learning process also play a crucial role. Future work could explore the impact of multimodal learning data, such as clickstream data [63] and response times [28], on learners' knowledge states to enable a more comprehensive modeling of the learning process. Additionally, learners' responses are often influenced by outlier behaviors, such as guessing or careless mistakes [64]. Investigating these anomalies could enhance the tracking of knowledge state evolution. Finally, exploring causal inference to understand the underlying reasons for learners' performance could help uncover causal relationships [65] that drive changes in knowledge states, thereby enabling more interpretable knowledge tracing. These directions hold significant potential for advancing the field in the future.

CRediT authorship contribution statement

Huali Yang: Writing – review & editing, Writing – original draft, Methodology. **Junjie Hu:** Writing – original draft, Visualization, Validation. **Jinjin Chen:** Writing – original draft, Validation, Methodology, Formal analysis. **Shengze Hu:** Writing – original draft, Validation, Methodology, Formal analysis. **Jing Geng:** Writing – original draft, Resources, Investigation. **Qiang Zhu:** Validation, Writing – review & editing. **Tao Huang:** Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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